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January 11, 1979
SUPERSEDING
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FEDERAL SPECIFICATION

BOXES, MICROSCOPE SLIDE, PLASTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two sizes of plastic microscope slide boxes suitable for accommodating 25 and 100 glass slides respectively.

1.2 Classification. The microscope slide boxes shall be of the following sizes as specified (see 6.1).

Size 1 - Large, 100 slides.
Size 2 - Small, 25 slides.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Federal Specifications

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard.

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Federal Standards

FED-STD-123	- Marking for Domestic Shipment (Civilian Agencies).
FED-STD-406	- Plastic: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, MO., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, DC.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Specification

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specification and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Size 1 and Size 2 boxes shall be fabricated of phenol-formaldehyde, polypropylene or polystyrene plastic and shall be brown, white, black or gray in color. Cover and body shall be the same color and shall be fabricated of the same material. The plastic materials shall have the physical properties listed in table I, when tested as specified in table V.

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TABLE I. Physical Properties.

Physical property	Phenolformaldehyde	Polypropylene	Polystyrene
Deflection temperature °F. under load (minimum)	290°	120°	160°
Izod impact strength (1/8 in. width). Foot pounds/inch of notch (minimum)	0.26	0.5	0.99
Tensile strength (1/8 in. width) pounds per square inch (psi) (minimum)	7000	--	3000
Tensile yield strength at 2 in. per minute testing speed, (psi) (minimum)	--	4500	--

3.2 Construction.

3.2.1 Size 1. Box shall be fabricated of plastic material and shall consist of a body and hinged cover which shall provide a flush fit on the top edge of the body wall. Phenolformaldehyde and polystyrene boxes shall have two hinges on one long side and a slide clasp or other suitable locking feature on the opposite long side to assure convenience and ease in opening and closing. Polypropylene boxes shall have one 15 cm (min.) length hinge in center of one long side and suitable clasp or other locking feature on the opposite long side. Hinges for phenolformaldehyde and polystyrene boxes shall be formed by metal pins driven into plastic hinge leaves which shall be integral with the cover and body or shall be secured to the cover and body by metal pins. Hinges and locking devices for polypropylene boxes shall be integral parts of the body and cover to form a one-piece box. Pins and other hardware shall be brass or steel (plated with nickel or chrome) or nickel.

3.2.1.1 Body. The interior of the body shall be fitted with three slide retainers molded integrally with the body. One slide retainer shall be along each of the two long sides and one shall be along the center long axis. Slide retainers shall have slots for 100 microscope slides in two rows of fity slides each. The two retainers along the sides of the box shall be slotted veritcally at equidistant intervals on the inner face of each retainer, and the center retainer shall be slotted vertically on both opposing faces in lines with the corresponding slots of the slide retainers.

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Both center and side retainers shall be reinforced to prevent cracking and splitting. A soft cork or suitable foam plastic material shall be firmly affixed to the inside bottom to act as a cushion for reduction of breakage and rattling of the slides. Outside bottom surface shall make four-point contact with a plane surface to prevent tilting and jiggling. Box shall be capable of accommodating one hundred glass microscope slides 25 millimeters (mm.) wide by 2 mm. thick by 73.5 mm. long as well as slides of the same width and thickness, but 77 mm. in length. Internal height of the box shall be such that the slides shall project above the top edge of the uncovered body by 6 ± 1 mm.

3.2.2 Size 2. Box shall consist of a body and a cover. Body and cover shall be so constructed that they form a flush fit when the box is closed. Cover shall be either removable or connected to the body by a hinge and shall have a suitable device to effect secure closure. Hinge of hinged box shall be 8.75 cm (min.) long in center of one of the long sides of the box.

3.2.2.1 Body. The interior of the body shall have two slide retainers which are integrally molded to the long walls. Retainers shall have slots for twenty five microscope slides. The two retainers shall be slotted vertically at equidistant intervals and each slot shall be in line with the corresponding slot of the opposite retainer. Each pair of slots shall be capable of accommodating one slide 25 mm. wide by 2 mm. thick by 73.5 mm. long as well as slides of the same width and thickness and 77 mm. in length. Internal height shall be such that the slides project over the top edge of the uncovered body by 6 ± 1 mm. The slides shall be retained in their proper slots, when the closed box is placed in any position. A cork or cork-like lining shall be permanently affixed to, and completely cover, the entire interior bottom except for the slide retainers and slots. The exterior bottom surface shall make four-point contact with a plane surface to prevent tilting.

3.3 Dimensions. Dimensions shall be as shown in table II.

TABLE II. Dimensions.

	Size 1	Size 2
Length	22.2 ± 1.3 cm.	13.9 ± 2.5 cm.
Width	16.5 ± 1.3 cm.	8.75 ± .6 cm.
Height (exclusive of four-point contact)	3.8 + .25 cm. - .4 cm.	3.1 ± .3 cm.
Wall thickness	1.5 min. (minimum)	1.5 min. (minimum)

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3.4 Marking.

3.4.1 Numbering and label, Size 1. A printed, numbered index label shall be firmly attached to the inside of the lid with corresponding numbers on the cork-type cushion on the inside bottom of the slide compartment to permit ready reference to the slides. The numbered index label shall cover the entire surface of the lid and shall consist of two rows of numbered lines from 1 through 50 in one row, and from 51 to 100 in the other row. Numbers designating each slide emplacement may be printed on top of the central divider in lieu of the numbering on the cork cushion. Numbers and lines shall be distinct and legible.

3.4.2 Numbering and label, Size 2. A printed, numbered index label shall be firmly attached to the inside of the lid with corresponding number on the cork lining on the bottom of the slide compartment to permit ready reference to the slides. The index label shall consist of one row of numbered lines, from 1 through 25, and shall cover the entire surface of the lid. Numbers and lines shall be distinct and legible. Numbers designating each slide emplacement may also be printed or molded adjacent to the upper edge of the slide partitions.

3.4.3 Marking for identification. Each box shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.5 Workmanship. Boxes shall be free from defects listed in table IV and any other defects which detract from appearance or impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table III.

TABLE III. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	I	1.0
Minor defects	I	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Plastic shall be sampled in accordance with FED-STD-406. When size of sample is not specified, at least two samples shall be taken from each batch of plastic.

4.3 Examination. Boxes shall be examined to determine compliance with all requirements contained in this specification. Examination shall include, but shall not be limited to, the defects listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major	
101	Box not free of cracks, chips, breaks, flash or sprues.
102	Box not free of warping and other distortion.
103	Box not free of sharp edges.
104	Cover not opened without difficulty.
105	Slides 25 mm. by 73.5--77 mm., not easily inserted or removed.
106	Slides not retained in place when box is held in any position with cover secured.
107	Cover cannot be completely closed with slides inserted in slots.
108	Box not free of rocking and wobbling when placed on a plane surface.
109	Inside bottom surface of slide compartment (exclusive of slots and retainers) not completely lined.

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TABLE IV. Classification of defects (continued).

Categories	Defects
Major	
110	Lining not properly secured.
111	Slots not properly aligned.
112	Slots not free of obstruction.
113	Index label not properly secured to inside surface of cover.
114	Index label not free of tears.
115	Index number missing, illegible or incomplete.
116	Index number does not correspond to slot number.
117	Index label not completely covering inside of lid.
Minor	
201	Box not free of scratches and mar marks.
202	Identification marking not complete, correct, permanent or legible.
203	Slot number or index number not legible.
204	Index label not free of wrinkles, smudges and extraneous marks.
205	Box not free of dirt and foreign matter.
206	Box not free of feathered edges and protrusions.

4.3.1 Dimensional examination. The boxes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.2 Packaging inspection. The inspection of the packaging, packing, and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Testing for plastic shall be conducted in accordance with table V.

TABLE V. Plastic testing.

Characteristic	FED-STD-406 Test Method Number
Deflection temperature	2011
Izod impact strength	1071
Tensile strength (1/8 in.)	1011
Tensile yield strength	1011

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

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5.1.1 Level A.

5.1.1.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Intermediate package. Six units of size 1 or twelve units of size 2 shall be packaged in box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Seventy-two units of size 1 or one hundred forty-four units of size 2 shall be packed into a exterior container; shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Seventy-two units of size 1 or one hundred forty-four units of size 2 shall be packed into a exterior container; shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following.

(a) Title, number, and date of this specification.
(b) National Stock Number (NSN).
(c) Size of box required (see 1.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
Defense Logistics Agency - DM

Project Number 6640-1061

Civil Agency Coordinating Activity:
GSA - FSS
VA - DMS
HEW - NIH

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